R54: actual RR fiber and four binary degree-20 evaluators

The complete normal-RR fiber is now identified directly by source matrices. Four closed source points have binary residue fields and complete function evaluators. Their orbit traces give a source-bound sign-minus null result.

Full source and inverse products

All 528 retained multiplication columns bind the unit, lambda, V and actual descended nu actions. At a^5+a^2+1=0 the idempotent selects the actual supersingular base point. The 16 monomials lambda^i*nu^j, i, j=0, 1, 2, 3, form a complete basis of that component. Both ranks are 16 and

```
K F = E; F K = I16; Mlambda K = K L; Mnu K = K N
```

Both descended equations vanish as full matrices; all border products pass. A changed basis entry is detected. Source: fiber54-01.

Actual normal RR algebra	Split fiber component	Primitive coordinate	Binary point evaluators
32 source generators	rank E = rank K = 16	w = nu / b	Four degree-20 moduli
528 multiplication columns	K F = E; F K = I	Degree 16 over F32	a = rho_i(w), all 32 values
Original nu/source bindings	Full multiplication products	Four quartic prime factors	128 full conversions

Explicit equations and larger residue fields

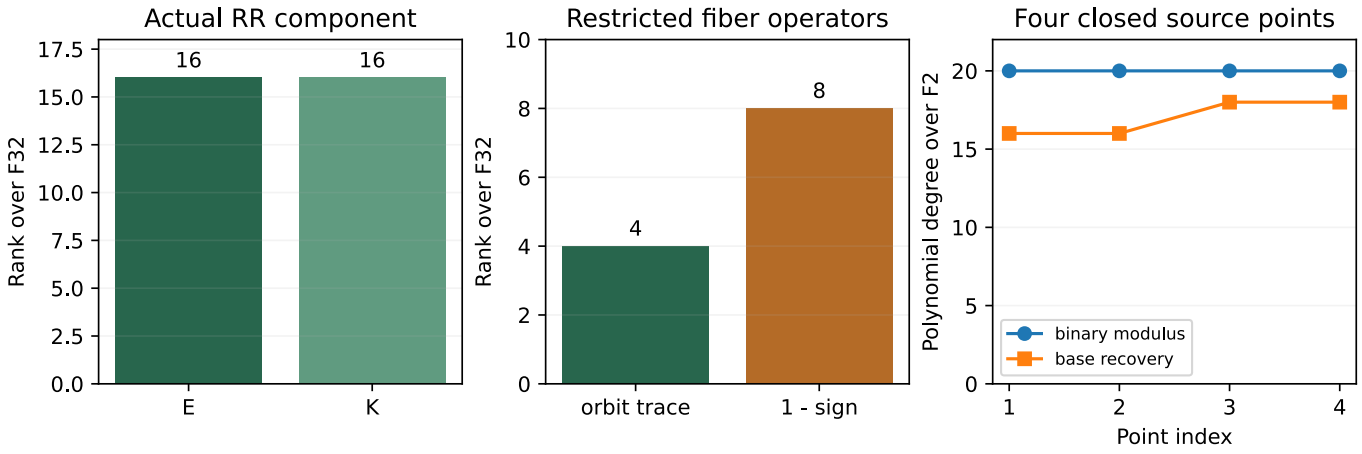
The actual border equations are

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lambda^4+b*lambda+b^2=0; nu^4+b^3*nu+b^4*(lambda+1)=0
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With w=nu/b, full ideal products give

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lambda=w^4+w+1; P=w^16+(1+b)*w^4+b*w+1+b+b^2
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P has four irreducible quartic factors over F32. Their five coefficient conjugates give four distinct irreducible degree-20 norms over F2. Every factor is checked independently by Frobenius/gcd tests. Each coupled point has all 32 RR basis values and all 528 multiplication products with full quotient witnesses: 2,112 products total.



All plotted values are exact native data. Sources: fiber54-01, frobfiber54-02, flatpoints54-01. Operator ranks are on this F32 fiber algebra, not global collector or Jacobian ranks. The complete anti-times-trace product is zero.

Complete binary evaluators and restricted obstruction

For each point the coupled binary ideal (a^5+a^2+1, g_i(a, w)) is identified by both full ideal products with (N_i(w), a+rho_i(w)). All four fields have degree 20 over F2. The rho degrees are16, 16, 18, 18. All128 RR values become binary polynomials of degree below20, with full source conversion products. The fresh checker independently replays all 2,112 RR products and checks unit and actual source-coordinate labels. Altered values are rejected.

Frobenius and the failed anti-projected direction

On the complete 16-dimensional F32 algebra the native matrices give

```
T=Frob32; T^2=S; T^4=I; S(w)=w+1
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The operator U=I+T+T^2+T^3 has rank 4, and I-S has rank 8. Their full product is zero. Thus these closed-point orbit traces are sign-invariant; they add no directions through the sign-minus projection. A changed trace is detected. No selected order-11 factor sign or global Jacobian action is inferred.

Lean and complete workflow ledger

Five authored implications cover basis injectivity, component recovery, exact range identification, order-four trace invariance and conditional anti-trace annihilation. Compilation and separate audit exit0. Lean also generated orbitTrace.eq_1: all six declarations audit with zero failing and zero trusted. Actual arrays, field arithmetic and normal-RR geometry are external premises.

Workflow	Status	Receipt
fiber-lean54-01	accepted	compile 0; audit 0; 6 declarations
fiber54-01	accepted	exit 0; 0 error lines
flatpoints54-01	accepted	exit 0; 0 error lines
frobfiber54-01	rejected	exit 0; 25 error lines
frobfiber54-02	accepted	exit 0; 0 error lines

Four workflows accepted, one rejected. The first primitive run used the reserved identifier degree; point checks were skipped and its late marker is rejected. The corrected attempt changes only local names. Immutable copied drivers and all 29 historical source/protocol byte pins are verified.

Native limits: 4 GiB virtual, 1200 s, one Singular/FLINT thread, seed54. Lean 4.34.1, mathlib d13f23b, -j1/-M4096, 1200 s separately for compile/audit, no virtual cap. Only native wall times are recorded; complete CPU times were not recorded. No accepted producer is repeated for publication. No comparative speed claim.

Next construction

Finish compressed global target checking through Frobenius powers and the QR/G, N syzygies, then the actual level-11/ degree-128 source map, F2 descent and intended 11 action. Test the larger A=0 quotient of dimension 80 and its possible CRT coordinate model before claiming that global quotient. Determine actual cusp/infinity poles before declaring a complete principal divisor. Keep proper polarization, full saturation, larger 112/991 and reverse-class lanes.

Sources: R50/R51 complete RR algebra, R52 actual nu action, R53 descended fiber; native linalg.lib; Stacks0BJF; standard finite-field Frobenius/gcd criterion. See CM.md, PROOF_MAP.md, protocol.json and results.json for premises and receipts.